

BRANDON, MB, Canada

WMO: 711400

Lat: 49.9033N

Lon: 99.9456W

Elev: 1343

StdP: 14.00

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-26.9	-22.4	-35.1	0.8	-26.5	-30.7	1.0	-22.2	29.3	11.6	25.3	11.4	8.8	280	0.698

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.1	86.0	69.2	82.8	67.2	79.8	65.4	72.0	82.1	69.7	79.7	67.6	76.8	10.8	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
68.8	111.1	78.3	66.2	101.4	75.0	64.0	93.9	72.8	36.7	82.1	34.6	80.0	32.8	76.9	83.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.2	21.8	19.0	-32.9	93.2	4.7	3.2	-36.3	95.5	-39.0	97.4	-41.7	99.2	-45.1	101.5
			-33.1	76.6	4.6	3.1	-36.5	78.9	-39.2	80.7	-41.8	82.4	-45.1	84.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	36.4	2.7	6.3	20.1	38.1	50.8	61.2	65.9	64.1	54.8	40.2	22.9	8.3		
	DBStd	24.93	14.82	13.83	13.92	10.80	8.49	6.46	5.44	6.15	7.94	8.88	11.93	13.78		
	HDD50	6557	1467	1222	928	374	94	4	0	0	40	322	812	1294		
	HDD65	10619	1932	1642	1392	807	446	148	56	91	316	768	1262	1759		
	CDD50	1607	0	0	1	16	118	339	492	438	184	19	0	0		
	CDD65	195	0	0	0	0	5	33	83	64	10	0	0	0		
	CDH74	2516	0	0	0	5	133	403	878	866	219	12	0	0		
	CDH80	636	0	0	0	0	28	89	206	265	46	2	0	0		
(14) Wind	WSAvg	9.3	9.5	9.4	9.5	10.1	10.6	9.2	7.8	8.1	9.0	9.7	9.5	9.2		
(15) Precipitation	PrecAvg	18.5	0.8	0.7	0.9	1.5	2.0	2.7	2.9	2.8	2.1	0.9	0.6	0.8		
	PrecMax	28.4	1.9	2.5	2.2	4.4	4.9	7.6	6.3	8.7	4.4	2.9	1.6	1.8		
	PrecMin	12.6	0.1	0.0	0.1	0.0	0.2	0.4	0.5	0.1	0.3	0.0	0.1	0.2		
	PrecStd	3.8	0.4	0.6	0.6	1.1	1.2	1.5	1.7	2.2	1.1	0.9	0.4	0.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	36.9	37.8	57.5	73.3	83.9	87.4	89.0	92.1	85.3	74.2	55.7	39.9		
		MCWB	34.0	34.5	46.5	54.8	61.7	69.4	72.6	70.3	65.8	57.8	45.4	35.2		
	2%	DB	32.9	34.0	48.0	66.6	77.6	82.2	85.0	86.4	79.9	65.9	48.1	34.6		
		MCWB	30.9	32.1	41.7	51.0	57.8	65.3	70.7	68.7	63.3	53.0	40.7	31.9		
	5%	DB	28.8	31.1	41.2	62.0	72.5	78.5	82.0	82.4	75.7	60.1	42.5	30.2		
		MCWB	27.3	29.7	37.2	48.5	55.6	63.4	69.0	66.9	61.3	49.8	37.9	28.6		
	10%	DB	24.5	27.0	36.8	56.3	67.6	75.2	79.0	78.8	70.6	54.8	38.0	26.9		
		MCWB	23.3	25.5	34.2	45.9	53.4	62.3	67.1	65.1	58.3	47.5	35.0	25.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	34.2	35.0	47.9	56.1	64.8	73.9	76.1	74.6	69.3	58.7	46.9	35.7		
		MCDB	36.6	37.2	56.4	69.8	77.5	84.3	84.4	85.3	79.6	72.6	54.2	39.1		
	2%	WB	31.1	32.5	41.9	52.4	60.3	68.7	72.7	71.0	65.5	54.5	41.9	32.0		
		MCDB	32.9	34.0	48.2	64.4	71.2	78.1	82.2	82.5	76.0	63.2	47.1	34.5		
	5%	WB	27.5	29.6	37.3	49.2	57.7	65.9	70.6	68.7	62.9	51.3	38.0	28.7		
		MCDB	28.9	31.1	40.9	60.2	68.7	74.6	79.9	79.5	72.3	58.2	41.9	30.1		
	10%	WB	23.3	25.4	34.4	46.4	54.9	63.7	68.4	66.5	60.0	48.1	35.1	25.7		
		MCDB	24.5	26.9	36.5	55.5	65.2	72.1	77.1	75.8	68.4	54.6	38.0	26.9		
(35) Mean Daily Temperature Range	MDBR		17.1	18.2	17.8	21.2	23.3	20.8	22.1	24.5	23.7	20.0	16.6	16.1		
	5% DB	MCDBR	19.8	17.9	20.6	31.1	32.3	27.0	26.3	30.7	32.5	30.4	23.9	19.3		
		MCWBR	18.4	16.1	15.0	18.1	16.4	13.9	14.2	15.5	17.6	18.6	17.2	17.2		
	5% WB	MCDBR	19.4	17.2	19.9	28.8	26.3	22.7	24.1	26.7	27.8	27.6	22.3	18.5		
		MCWBR	18.0	15.5	14.9	17.4	14.8	13.0	14.2	14.6	16.3	17.5	16.5	16.6		
(40) Clear-Sky Solar Irradiance	taub		0.260	0.265	0.305	0.344	0.357	0.367	0.381	0.383	0.347	0.290	0.262	0.235		
	taud		2.271	2.283	2.308	2.348	2.373	2.384	2.352	2.340	2.418	2.547	2.472	2.378		
	Ebn at Noon		251	281	287	286	285	282	276	270	269	272	254	250		
	Edh at Noon		22	28	33	36	36	36	37	36	30	22	18	17		
(44) All-Sky Solar Radiation	RadAvg		453	826	1244	1599	1792	1892	2013	1647	1205	725	404	323		
	RadStd		36	62	96	110	143	131	91	124	94	91	56	34		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (21 neighbors)		N/A	N/A	N/A	N/A	N/A	-1.19	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page